

NexusFinance — Technical Architecture

Digital Lending Platform for Cambodia

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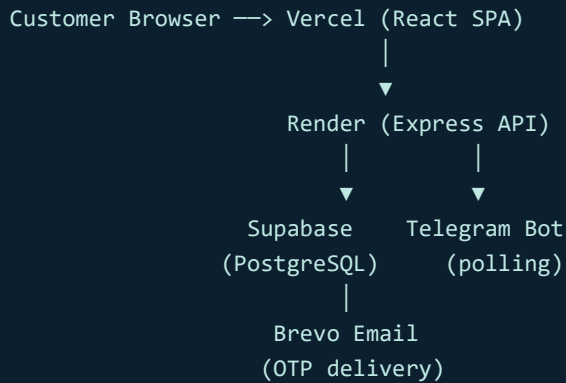
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Architecture • Tech Stack • Deployment • DevOps

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1. System Architecture



2. Technology Stack

Layer	Technology	Purpose
Frontend	React 18, TypeScript, Vite, Tailwind CSS	User interface
Backend	Node.js, Express, TypeScript	REST API server
Database	Supabase (PostgreSQL)	Data storage
Auth	JWT + bcrypt + Brevo OTP	Authentication
Hosting	Vercel + Render	Deployment
Payments	KHQR, ABA PayWay	Payment processing
Notifications	Telegram Bot API, Brevo	Messaging
Mobile	Expo SDK 57	Android app

3. Project Structure

```
nexusfinance/
├── src/
│   ├── components/    # React frontend
│   │   ├── App.tsx    # UI components
│   │   ├── api.ts     # Main app with routing
│   │   └── index.css  # API client
│   └── index.ts       # Design tokens
├── server/            # Express backend
└── index.ts           # API routes + server start
```

```

|   ├── db.ts           # Supabase client
|   ├── bot.ts          # Telegram bot
|   ├── brevo.ts        # Brevo email API
|   ├── otp.ts          # OTP generation/verification
|   ├── khqr.ts         # KHQR QR generation
|   └── sms.ts          # Twilio SMS
|── mobile/             # Expo Android app
|── supabase/migrations/ # Database migrations
|── vercel.json         # Vercel rewrites to Render
|── package.json

```

4. Database Schema

4.1 Core Tables

Table	Key Columns	Purpose
nexus_users	id, name, email, password, role, phone, telegram_chat_id, otp_*	User accounts
nexus_loans	id, user_id, amount, type, status, duration_months, interest_rate	Loan applications
nexus_transactions	id, user_id, loan_id, amount, type, status	Payment transactions
nexus_audit_logs	id, action, description, user_id, metadata, created_at	Audit trail
nexus_tasks	id, user_id, title, status, priority	KYC and review tasks

5. Frontend Architecture

- Single-page application (SPA) with React 18 and TypeScript
- Three portals: Customer, Loan Officer, Super Admin
- Lazy-loaded route components for performance
- Design tokens in CSS custom properties (dark/light mode)
- Vercel proxies /api/* to the Render backend

6. Backend Architecture

- Express.js REST API with TypeScript

- JWT authentication middleware with role checks
- Rate limiting on auth endpoints
- Cron job for daily payment reminders (9 AM Cambodia time)
- Telegram bot starts automatically with the server

7. Deployment Pipeline

Component	Trigger	URL
Frontend (Vercel)	Git push to main	https://nexusfinancefintech.vercel.app
Backend (Render)	Git push to main (auto)	https://nexusfinance-lof3.onrender.com
Database (Supabase)	Migration files	https://jpltdxlljqcobjuhfvb.supabase.co

8. Environment Variables

All secrets are managed via environment variables on Render. Key variables:

Variable	Purpose
SUPABASE_URL	Supabase project URL
SUPABASE_SERVICE_ROLE	Server-side Supabase access key
BREVO_API_KEY	Brevo transactional email API key
TELEGRAM_BOT_TOKEN	Telegram bot authentication token
TELEGRAM_ADMIN_ID	Admin Telegram user ID
JWT_SECRET	Token signing secret
PAYWAY_MERCHANT_ID	ABA PayWay merchant ID
PAYWAY_API_KEY	ABA PayWay API key
PORT	Server listening port (set by Render)

9. Local Development

```
git clone https://github.com/ndxdigitalsupport/nexusfinance.git
cd nexusfinance
npm install
cp .env.example .env # fill in your values
npm run dev
```

Frontend runs on <http://localhost:5173>, backend on <http://localhost:3001>.